

Introduction.

from Khovanov homology to exotic TR^4 .

Khovanov homology $\xrightarrow{\text{Step 1}}$ s-invariant $\xrightarrow{\text{Step 2}}$ slice knot $\xrightarrow{\text{Step 3}}$ exotic TR^4
(Lee)

Step 1: take the q-degree. denoting it $s(K)$

Step 2: use the property of s-invariant (like a functor)

Lee gave the thm that

$$|s(K)| \leq 2g_*(K).$$

$\downarrow$ $\downarrow$
s-invariant. smooth slice genus
of a knot.

Step 3: There is a famous **Thm**:

$\exists K \subset S^3$, K topologically slice but not smoothly slice. $\Rightarrow \exists$ large exotic TR^4 .

• Definitions.

firstly, we give a proof of step 3, which is relatively independent.

• Def. slice knot. slice genus. topo slice and smooth slice. Seifert surface.

• slice knot (smooth. topo)

a knot $K \subset S^3 = \partial D^4$ smooth slice if it "bounds a smooth disk in D^4 " (flat topo)

by "bounds a smooth disk" we mean $\exists$ smooth embedding $D^2 \hookrightarrow D^4$ (flat topo).
s.t. $\partial D^2 = K$.

• flat topo embedding: $D^2 \hookrightarrow D^4$ a continuous embedding
& its tubular neighborhood $D^2 \times D^2 \hookrightarrow D^4$ also a continuous embedding.

• topo embedding is not enough: every K has a topo disk, just take the cone.

• slice genus (smooth)

• a smooth slice knot has slice genus zero.

• genus of any compact surface $\left\{ \begin{array}{l} \text{closed: defined.} \\ \text{with boundary: cap the boundary} \end{array} \right.$

• genus of Seifert surface.

$$K \subset S^3$$

$$\Sigma \hookrightarrow S^3 \text{ s.t.}$$

$$\partial \Sigma = K$$

$\left\{ \begin{array}{l} \text{Seifert surface. } \Sigma \\ \text{genus of Seifert: use Euler charac number} \end{array} \right.$

$\downarrow$
can be computed.

definition: least genus of

$$X(\Sigma) = 2 - 2g - b \quad \left\{ \begin{array}{l} \rightarrow 1 \text{ for knot} \\ \Rightarrow g \checkmark \end{array} \right.$$

$$= s - c$$

$\left. \begin{array}{c} \} \\ \} \end{array} \right\}$

Seifert crossings
circles points.

- slice genus (smooth)

- def. $\exists \hookrightarrow D^4$, $K \hookrightarrow S^3$ s.t. $\partial \exists = K$.

slice genus of K , denoting $g_*(K) := \min \{g(\exists) \mid \exists \text{ above}\}$.

- hard to compute.

Def. exotic $\mathbb{R}^4$

- exotic.

topo $\mathbb{R}^4$ but not smooth $\mathbb{R}^4$.

i.e. $\forall f: X \rightarrow \mathbb{R}^4$ homeo. f not diffeo.

- large.

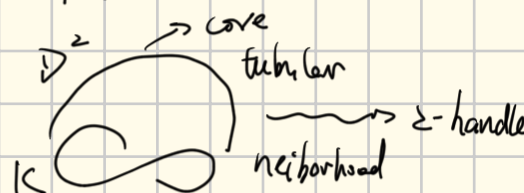
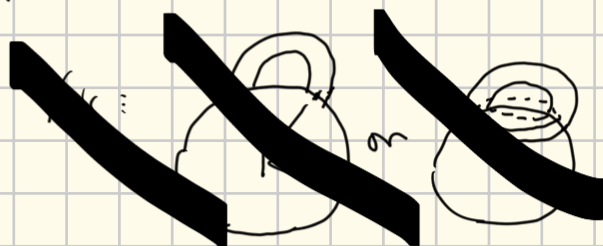
small exotic $\mathbb{R}^4$: X exotic $\mathbb{R}^4$ s.t. $\exists U \subset \mathbb{R}^4$
 $X \hookrightarrow U$ (essential)

large = not small.

Proof of the thm in Step 3.

i.e. $\exists K$ topo slice, not smooth slice $\Rightarrow \exists$ exotic $\mathbb{R}^4$ which is large.

- Def. $X_K = D^4$ attach 2-handle along $K \subset \partial D^4 = S^3$.



this direction is trivial.

- lemma 1. K topo slice $\Rightarrow X_K \xrightarrow{\text{topo}} \mathbb{R}^4 (S^4)$.

pf: $S^4 = D^4 \cup_{\partial D^4} D^4$
 $\uparrow \quad \uparrow$
 $(K \hookrightarrow D^4) \quad \text{2-handle.}$

pf of a smooth version
 is similar

i.e. $K \hookrightarrow D^4$ is one copy = obviously

2-handle can be put into another copy = core $\hookrightarrow D^4$
 attaching along S^3 .

$\Rightarrow X_K = ((K \hookrightarrow D^4) \cup_K \text{2-handle}) \hookrightarrow S^4$.

Fact 2. $\forall$ connected, non-cpt 4-mfld X , $\exists$ a smooth struct.

pf of the thm.

K topo slice $\Rightarrow X_K \xrightarrow{\text{topo}} \mathbb{R}^4$.

hence $\mathbb{R}^4 = \mathbb{R}^4 - \text{int}(X_K) \sqcup X_K$
 $\swarrow \searrow$
 connected non-cpt 4-mfld. $\xrightarrow{\partial X_K}$ closed 3-mfld $\xrightarrow{\text{in } \overline{\mathbb{R}^4}}$ has a inherent smooth structure

connected: 挖掉 0-handle+2-handle
 等于先挖掉 0-handle 再沿着 boundary
 挖掉 2-handle.

而沿着 boundary 挖 2-handle 等同于
 沿着 boundary 粘 1-handle

hence $\mathbb{R}^4 - \text{int}(X_K)$ a smooth mfld. $\xrightarrow{\partial_1 \text{ smooth}}$
 X_K a smooth mfld. $\xrightarrow{\partial_2 \text{ smooth}}$
 $\left. \begin{array}{l} \text{boundary: } \partial X_K \\ \partial_1 \text{ " } \partial_2 \end{array} \right\}$ 3-mfld

ie they are attached by some homeo. $\phi: \partial_1 \rightarrow \partial_2$

$\partial_i = \partial X_K$ a 3-mfld. has $\hat{=}$ unique smooth structure

hence ϕ not only homeo, but also a diffeo!

attaching the two smooth mfld along boundary
 we obtain a new mfld $M := M_1 \cup_{\phi} M_2$
 which is homeo. to $\mathbb{R}^4$.

M not a smooth $\mathbb{R}^4$.

if M a smooth $\mathbb{R}^4 \Rightarrow X_K \xrightarrow{\text{smooth}} \mathbb{R}^4$

a contradiction!

hence M is a exotic $\mathbb{R}^4$.

M not small.
 if small $X_K \xrightarrow{\text{smooth}} M \hookrightarrow \mathbb{R}^4$ 矛盾! $\square$